

Review Article

A survey on evaluation of blockchain-based agricultural traceability

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ABSTRACT

The integration of blockchain technology in agricultural traceability has shown immense potential, yet its widespread adoption faces significant roadblocks. Using bulk product traceability as a foundational reference, this paper presents a comprehensive evaluation framework for Blockchain-based Agricultural Traceability. The framework accentuates product identification and data traceability across the supply chain, addressing traceability disconnections caused by bulk product blending. It dives into depth levels from adoption decision-making to system design, development, and deployment, emphasizing the critical aspects of traceability compliance and standardization. As a result, we identified the obstacles to adopting agricultural digital traceability and pave the pathway to traceability system deployment. We examined the barriers to implementing digital traceability of agricultural products, taking the Australian grain supply chain as an example. Our findings reveal that lack of standardization and participation barriers are the primary challenges in implementing digital traceability for agricultural products. Our paper offers insights and recommendations for researchers, industry practitioners, and business owners to overcome these challenges and enable digital traceability of agricultural products in global supply chains.

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1. Introduction

The agriculture sector is under growing pressure to improve the traceability of its products, with increasing consumer demand for transparency and accountability in food supply chains. Digital traceability provides a clear and secure record of a product’s journey from farm to consumer. This helps to ensure the authenticity and quality of the product, improve supply chain transparency and efficiency, and enhance food safety and regulatory compliance (Mirabelli and Solina, 2020; Cao et al., 2021). Also, digital traceability supports auto transaction, since it provides a comprehensive, secure, and accessible record of a product’s journey, including information on its origin, movement, and handling. This information can be integrated with trade systems, such as electronic certification, shipping, and customs systems, to streamline trade processes and reduce the need for manual intervention, which enables faster, more efficient, and more secure trade transactions.

Blockchain has made a big difference of agricultural traceability by providing a shared platform for tracking agricultural products throughout the supply chain, offering a digital trust mechanism equivalent to how a third party, like a bank, acts as a witness or guarantor in real-world trade transactions. Unlike traditional systems, blockchain is a shared digital record that safeguards information from tampering, which allows everyone involved to trust the data. It simplifies the process of tracing where products come from, how they have been handled, and who has been involved in the process. This enhanced visibility has introduced a new level of trust and accountability in tracking agricultural products with ownership changes.

The adoption of blockchain for agricultural product traceability (Torky and Hassanein, 2020) has been limited, particularly among small and medium-sized enterprises (SMEs), as they are often hesitant to participate in blockchain traceability. The main obstacles include (1) lack of adoption pathway: SMEs often struggle with understanding the control points in the adoption process, and may not know how to assess the performance of their current implementation. This lack of understanding can create uncertainty and hesitation about whether to invest in new practices; (2) operation complexity: While the blockchain itself is trustworthy, implementing traceability requires coordinated actions from all stakeholders across the entire supply chain (Thakur et al., 2020). As an individual company, managing and ensuring the cooperation of every party involved in the supply chain can be almost impossible; and (3) cost constrain: For high-value agricultural products like tea or organic food, there is a demand for traceability. However, for products characterized by inherently low profit margins may encounter difficulties in affording blockchain solutions, whether they are consortium blockchains or public blockchains.

Addressing the above gaps is crucial to accelerate the adoption of new practices in the agriculture industry. To evaluate traceability techniques, we delve into the case of bulk agricultural product

traceability, which presents unique challenges and insights for industry implementation. Bulk agricultural products also called bulk commodity, are shipped loose or in mass/aggregate, which are transported in large volumes and confined by large containers or transportation units (Thakur, 2010). The range of agricultural products include Barley, Corn, Soya, Sunflower Seeds, Rapeseed, Wheat, Maize, Cotton, Canola, and Soybeans etc. Across supply chain from farm to market/export, bulk agricultural product, e.g., grain, from different storage bins (i.e., different quality) is likely blended to meet the customer specifications for quality and to maximize the elevator’s profit (Thakur and Hurburgh, 2009a). The blending can lead to traceability disconnection and increase the likelihood of errors, such as mislabeling or misrepresentation of product origins, quality, or safety attributes. Moreover, the agricultural supply chain involves numerous intermediaries, such as traders, processors, and distributors, who may use different systems or standards for data management, resulting in data fragmentation and inconsistent information. Therefore, performing traceability on bulk agricultural products presents technical challenges due to the nature of these products and the complex supply chain processes they undergo.

1.1. Related survey and contributions of the paper

In literature, several surveys focusing on different aspects of agricultural tractability have been conducted during the past few years including internet of things (IoT), blockchain, with focus on method study, system design, and security. A few research papers address the combined perspective of IoT and blockchain and their smart agriculture applications. For instance, Friha et al. (2021) presents a comprehensive review of emerging technologies for the IoT-based smart agriculture, which covers categories of smart monitoring, smart water management, agro-chemicals applications, disease management, smart harvesting and supply chain management, making it a valuable resource resembling a handbook or dictionary of smart agriculture technology categories. It is particularly useful for implementing digital traceability in agricultural products.

Table 1 summarizes the most recent surveys on agriculture traceability from the viewpoint of product identification, blockchain, supply chain, traceability standardization and assessment. It is important to note that these surveys have certain limitations when it comes to technology adoption. They primarily provide a overview of technologies through categorization, but not with a focus on traceability evaluation. Thus, it lacks precise measurements or step-by-step instructions for businesses to follow in integrating these techniques. Furthermore, the review does not delve into the important aspects of compliance and

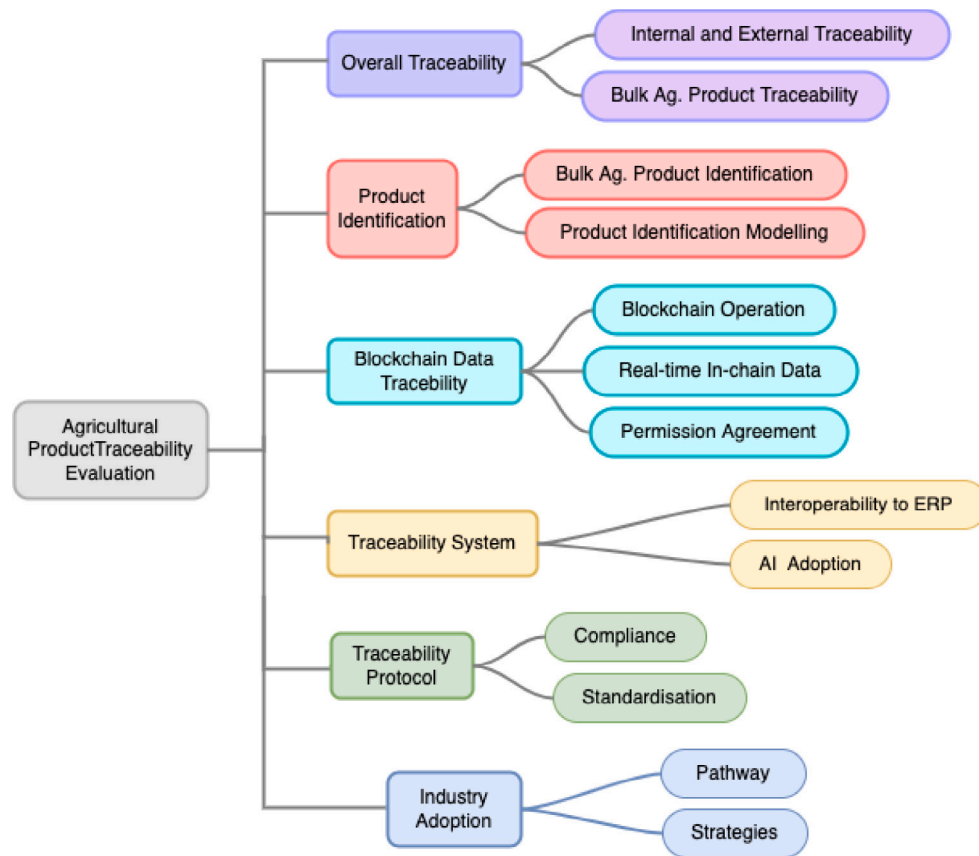


Fig. 1. The outline of agricultural traceability evaluation.

Table 1
Related surveys on agricultural product traceability.

Survey	Identification	Blockchain	Supply Chain	Compliance	Assessment	Year
Muhammed et al. (2024)	✓	✓	×	×	●	2024
Pandey et al. (2022)	×	✓	×	×	●	2022
Tanwar et al. (2022)	×	✓	✓	×	●	2022
Stranieri et al. (2021)	×	✓	×	✓	●	2021
Li et al. (2021b)	×	✓	✓	✓	×	2021
Friha et al. (2021)	✓	✓	✓	×	×	2021
Demestichas et al. (2020)	×	✓	×	×	×	2020
Mirabelli and Solina (2020)	×	✓	✓	×	×	2020
Feng et al. (2020)	✓	✓	✓	✓	×	2020
Antonucci et al. (2019)	×	✓	×	×	×	2019

(✓): Supported; (×): Not supported; (●): Partially supported.

standardization, which are currently under development, but future will determine the success of digital traceability.

1.2. Contribution of this paper

The primary contributions of this paper are summarized as:

- Development of a robust evaluation framework that gauges the performance, impact, effectiveness and entry to blockchain-based agricultural traceability.
- Investigation into the traceability of bulk agricultural products, addressing the challenges of inconsistent data capture, and offering a technical reference for real-world evaluations of agricultural product traceability.
- Insights into traceability compliance and standardization prerequisites for the widespread adoption of blockchain-based agricultural product traceability.

- Examination of the barriers to industry adoption and proposition of a roadmap for businesses considering a shift to blockchain-based traceability.

The rest of the paper is organized as follows: Section 2 introduces the general architect of agricultural traceability. Section 3 discusses the overall traceability performance evaluation. Section 4 examines the methods used in assessing agricultural product identification, where we delve into the intricacies of bulk product identification and highlight the requirement of data modeling for consistent identification. Section 5 presents the performance metrics on blockchain operation for data traceability. Section 6 delves into blockchain-integrated agricultural traceability systems, detailing the interplay of digital technologies and their seamless integration with ERP systems. In Section 7, we explore the pivotal role of compliance and standardization in agricultural traceability, emphasizing the significance of unified protocols and guidelines for seamless interoperability and transparency across supply chains. Section 8 designs the pathway of industry adoption and promotes the strategies to undertaken by bossiness owner and

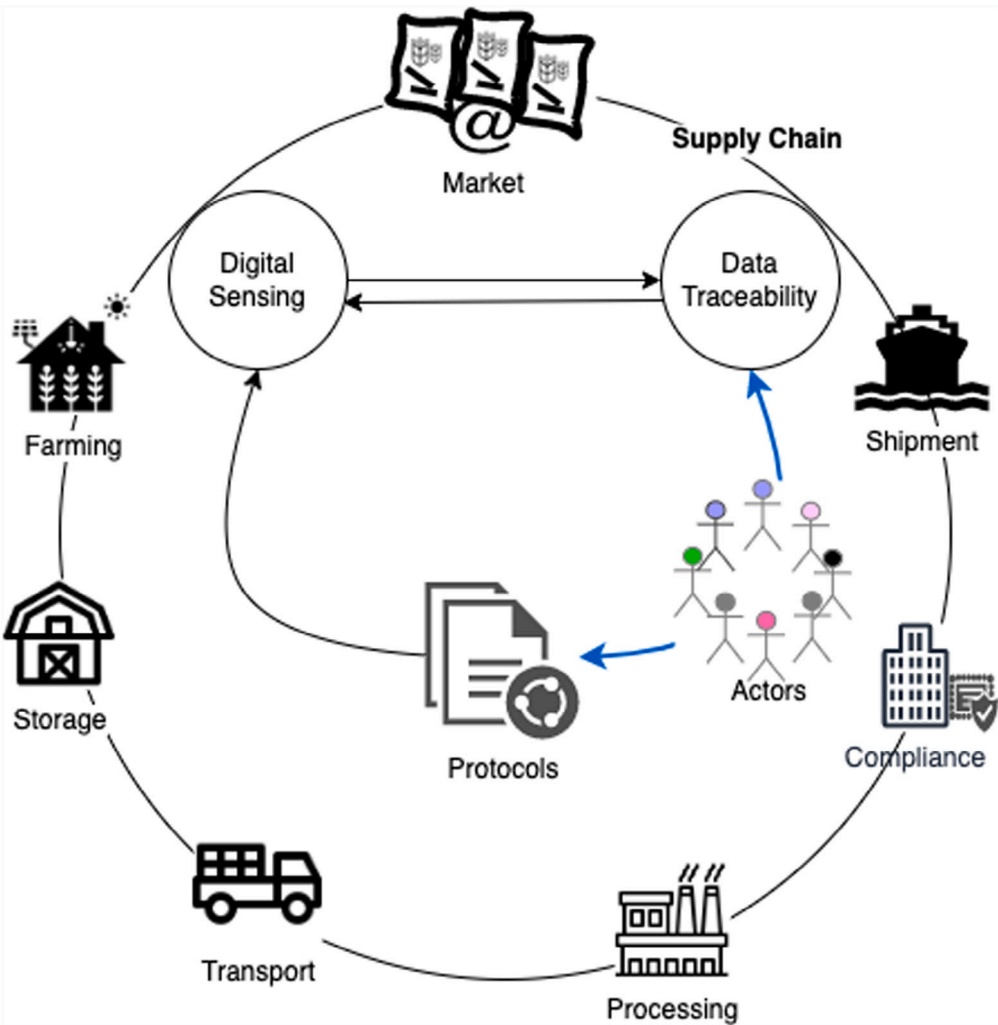


Fig. 2. The overall architect of agricultural product traceability, where the traceability process from farming to market is modeled as three key components including digital sensing, data traceability, and protocols guiding interactions between actors across the supply chain.

government to facilitate the industry adoption. Finally, in Section 10, this paper concludes with the presentation of ideas for future work. To help readers understand this paper, acronyms found in this paper are shown in Table 2, and the outline of our agricultural traceability evaluation is presented in Fig. 1).

Fig. 2 presents the outline of agricultural traceability evaluation, which illustrates the logic and methodology of the review. The methodology follows a structured approach across two key dimensions. The first dimension focuses on discovering the elements of agricultural traceability, including product identification, data traceability, and traceability protocols. This involves examining how each of these elements contributes to the overall traceability system. The second dimension delves into Blockchain-based Agricultural Traceability (BAT), exploring components such as blockchain operation, private agreements, and in-chain data management. Furthermore, BAT is investigated as a comprehensive system, covering its design, development, deployment, and adoption stages. This approach ensures that both technical and managerial aspects are thoroughly examined.

2. Background

2.1. Digital traceability architect

A typical digital traceability system can be generalized as three connected components: (1) digital sensing for capturing data at different stages of the supply chain; (2) data traceability servers for

Table 2
Acronyms used in this survey.

Acronym	Description
IoT	Internet of things
AI	Artificial intelligence.
UM	Uniformly mixed
FIFO	First in first out
RFID	Radio frequency identification
QR code	Quick response code
QoS	Quality of service
PoW	Proof of Work
PoS	Proof of Stake
ERP	Enterprise resource planning
TS	Traceability system
TRU	Traceable resource unit
SME	small and medium-sized enterprises

supply chain actors to trace products over time and location; and (3) a set of protocols/standards to be followed to perform traceability via (1) and (2). In practice, the reliability of our traceability systems directly influences the reduction of duplication and costs associated with traceability implementation. Fig. 2 gives the overall architect of agricultural product traceability.

The existing traceability systems adopt either of the two architectures: centralized architecture or distributed architecture. Centralized

traceability system is managed and maintained by an authoritative third party. It may suffer the single node attack and has higher risk of data tampering and information disclosure (Lin et al., 2019). Distributed architecture is where the data are stored in multiple locations or nodes. Each node in the network has a copy of the data, and each node can verify the authenticity of the data. This architecture is commonly used in blockchain-based traceability systems (Caro et al., 2018; Lucena et al., 2018; Gunasekera and Valenzuela, 2020; Kumar and Iyengar, 2017; Salah et al., 2019; Lin et al., 2019), where the data are distributed across a network of nodes, and transactions are verified through a consensus mechanism.

2.2. Agricultural product supply chain

A typical supply chain of agricultural product consists of the following stages: production, storage, processing, transport, and trade. Production also named as farming, describes the process starting from how the farmers buy seeds from a seed company, to preparation of soil, sowing, irrigation, fertilizing, harvesting and storage. Storage stage means the operation of farm product warehousing and storage facilities to the highest standards and are audited and checked regularly to ensure all certification requirements are being met. With reference to Australian grain supply chain https://www.graintrade.org.au/sites/default/files/file/Codes/GTA_103199_CODE_BRO_WEB.pdf, typical transport of agriculture product includes the option of via road, rail and coastal shipping. Transport stage traceability includes the operation of packing, in-loading and ex-loading, and transportation to ensure all standard requirements are being met. For bulk product, packing is one process of transport, since improper packing can cause distortion and compromise the containers integrity resulting in ‘bulging’ of the unit and spillage of contents through container doors. Processing involves physico-chemical and/or microbiological processes such as heating, cooling, concentration, and pasteurization. Trade involves change ownership of product, which consists of retail to terminal consumer, mid trade and export.

The actor of the above supply chain includes: Caro et al. (2018) (a) provider: providers of raw materials, such as seeds and nutrients, but also pesticides, chemicals, etc; (b) producer: usually the farmer e.g., the responsible of the actions from seeding/planting to harvesting; (c) processor: this actor may perform various actions, from simple packaging to more complex processes (e.g., pressing of the olives); (d) distributor: this actor is responsible of moving the output of the processor (e.g., the product) from processor’s site to retailers; (e) retailer: this actor is responsible of selling the products, representing it either small local stores or big supermarkets; (f) consumer: the final element of the chain; and (g) regulator: the actor responsible for overseeing the chain and enforcing laws, regulations, and standards.

In the case of product for export, before distributor the supply chain has additionally the following actors: (h) Exporter: The actor that purchases agricultural products from producers and processors, and prepare and transport them for export to international markets; and (i) Transporter: The actor that arranges the logistics of transporting agricultural products from the point of origin to the destination port, then to the overseas market, including customs clearance and documentation.

It is important to note that the sequence of actors may vary depending on the specific product and location, and additional actors such as brokers, traders, and government agencies may also be involved in the supply chain for agricultural product export.

2.3. Internal and external traceability

Traceability throughout the supply chain consists of multiple stages of internal and external traceability (Qian et al., 2017). External traceability, often referred to as chain traceability, encompasses the sharing of product information among various stakeholders within a supply

chain. It is important to note that external traceability frequently involves tracking the transfer of product ownership along the chain. This process is essential for regulatory compliance, as it enables authorities to trace the origin and journey of products, which can be crucial in cases of recalls or safety concerns.

Internal traceability (Thakur and Hurburgh, 2009a; Lucena et al., 2018), also known as intrinsic traceability, involves maintaining comprehensive records of agricultural products within a designated operation, company, or production facility. This meticulous approach encompasses the continuous monitoring of processes, stringent quality control, effective inventory management, precise batch and lot tracking, resource optimization, and adherence to regulatory compliance. Internal traceability ensures the efficient handling of agricultural products, upholds stringent quality standards, optimizes the allocation of resources, and guarantees compliance with crucial regulations, all without involving any changes in product ownership.

The full-capacity traceability includes both external supply chain traceability as well as internal traceability. Internal should be linked with external, so that when demanded, the information can be provided for real-time product quality inspection. Fig. 3 illustrates the full traceability of grain supply chain from farm to export, where each stage identifies an internal traceability in multiple processes. For instance of grain quality assurance, farming stage can be further investigated on wheat seed, chisel plough, ground rig, fertilizing, sowing, ground rig, and harvesting.

3. Overall traceability assessment

Traceability helps create a snapshot to identify coverage gaps and gives us visibility throughout the development lifecycle, thus accelerating the development and reducing waste of resources like time and effort. For a traceability system designing, the following assessments (Sahin et al., 2002) should be considered to ensure quality implementation.

- The degree of detail of information about items monitored by the traceability system.
- The degree of automation (manual, semi-automatical, automatic) of the item identification and data collection process.
- The percentage of supply chain participants involved for external traceability
- The percentage of supply chain stages/processes covered in the life cycle of the item product.
- The flexibility of tracing back (supplier traceability) and tracking forward (client traceability).
- The immutability of data tracking across different stakeholders.
- The data security of the traceability system.
- The speed and throughput of the system response.

In operating a traceability system, detailed technical performance assessments are applied. This includes precision, breadth, depth, and granularity, and other indicators which have been used to measure the effectiveness of traceability (Stranieri et al., 2021; Zheng et al., 2023). In addition, purity was used in horticultural pack house processing transformations, and capability, rapidity, and accuracy has been used in fish processing plants (Qian et al., 2017).

Note that for bulk agricultural product traceability, the degree of consistent product identification and labeling throughout the supply chain is an alternative key criterion. This is similar to traceability processed food faces because of the variety of raw materials, batch mixing, and resource transformation (Comba et al., 2013). In this sense, traceability is evaluated based on recall effects, TRUs (traceable resource units), and comprehensive granularity (Qian et al., 2022b; Wang et al., 2017).

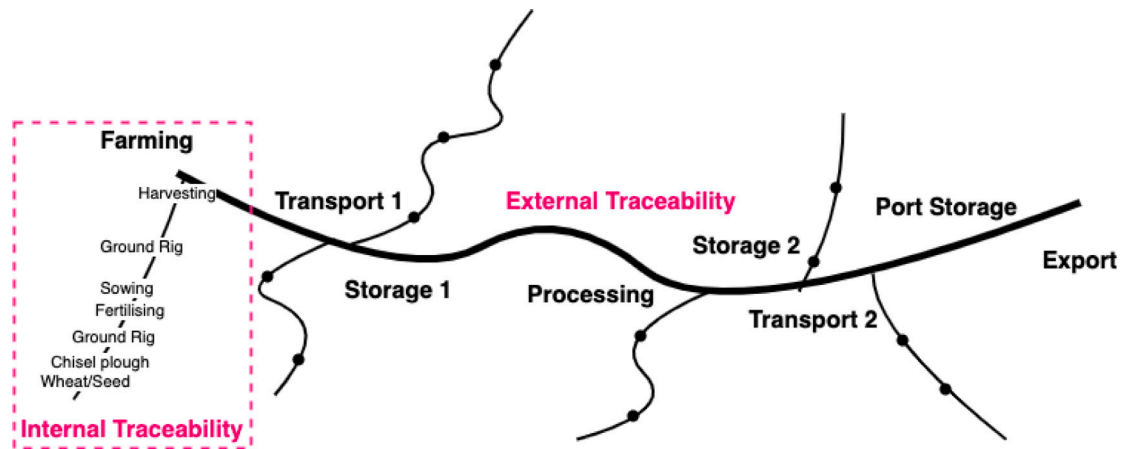


Fig. 3. The illustration of agricultural product traceability structure: internal and external traceability.

4. Product identification

External traceability requires all traceable resource units (TRU) to be uniquely identified, and information to be shared between all affected supply chain actors. However, in practical implementation, discrepancies may arise due to varying contexts such as differences in technological infrastructure, data entry procedures, or regional regulatory standards, leading to potential errors (Badia-Melis et al., 2015) or inconsistencies (Bosona and Gebresenbet, 2013) in the traceability process.

4.1. Accessing product identification

For effective traceability, a paramount factor lies in the precise and consistent identification of agricultural products within the traceability system. To comprehensively assess the performance of product identification, typical measurements include:

- Accuracy: with minimal errors or false positives (Bosona and Gebresenbet, 2013).
- Speed: fast enough to meet the requirements of the supply chain (Ellahi et al., 2023).
- Scalability: be able to handle large volumes of data and be scalable as the business grows (Fan et al., 2020b).
- Reliability: be reliable and available at all times, with minimal downtime or disruptions (Lin, 2022).
- Security: have robust security measures in place to protect against unauthorized access or tampering (Fan et al., 2020b).
- Integration: be easy to integrate with other systems and tools in the supply chain (Ellahi et al., 2023).
- Cost-effectiveness: be cost-effective, with a reasonable return on investment (Fan et al., 2020b).

For each of measurement here and after, please follow the references given to find out how the measurement can be quantitatively calculated/ modeled in practice.

4.2. Inconsistent product identification

In the case of bulk agricultural product, product is required to be traced from the field to the consumer by lot through unique codes. A code could be a number, number-letter combination, or some other designation that is unique to the lot. Traceability thus can be implemented by establishing lots with each distinct lot traced separately. However with types of bulk agricultural products, it is very difficult to associate any kind of label, marker or identifier that could directly identify the different lots (Orzechowski, 2019). Even if the unique lot code is

identified on container or storage bin, the bulk agricultural product lots would be commingled in the context of (1) a trade where product lots are blended to meet consumer specifications and to maximize the profit, which was referred to as uniform-mixing (UM) in Comba et al. (2013); and (2) product transport and storage, the stored agricultural product is drawn from a container to be delivered to a production station or to a new storage container. As a result of such commingling, lot identity is not maintained, an outbound shipment to a customer likely contains products from many different sources. In this sense, tracing the source of a problem or tracking other affected shipments or storage would be nearly impossible, despite the use of advanced technologies such as RFID, barcodes, QR codes, and IoT.

Table 3 presents a range of identification technologies documented in the literature, which are employed for various agricultural products to facilitate data traceability. These technologies span from advanced IoT-enabled containers with comprehensive sensor arrays, as seen for soybeans, to more basic approaches like tags used for rice. It is significant to note that for the majority of these products, no identification modeling is adopted. This absence can potentially lead to inconsistencies, negatively impacting the effectiveness of traceability.

4.3. Product identification modeling

As a mitigate solution, fuzzy traceability was proposed to approximate the amount in a way to keep the lot identity for in-bounding. Liang et al. (2019) developed an automatic separation and identification equipment for grain tracing systems based on grain tracers with QR codes. However, this solution is only suitable for the management of internal traceability, i.e. during the production processes within a company.

For chain tractability, we can follow Comba et al. (2013) and Qian et al. (2017) the rule of FIFO (first in first out) as in Fig. 4 to record and link up the identity of every stage of the supply chain as Fig. 5. This requires the identification approach to be able to add digits constantly or combining multiple identification approaches. For example, establishing the relationship between the QR code affixed to the small package and the RFID pasted in the storage bin was an important step for supply chain traceability. After affixing the QR code label, the next step was to form a package bin by loading a number of small packages of flour. The disadvantage of this solution is, it supports trace forward, but not tracking backward.

Alternatively, under the condition that the product delivered to a compartment is instantaneously and uniformly mixed (UM) (Comba et al., 2013). By accurately recording the composition of the input flows and the storage mechanism in the UM compartment, we can track the movement of the product through the supply chain and ensure that it is being handled in accordance with safety and quality standards.

Table 3
Identification technologies and traceability modeling for 11 bulk agricultural products.

Agricultural product	Identification technology	Identification modeling	Ref.
Soybean	IoT-enabled containers and packages equipped with sensors, cameras, GPS locator and 4G	No	Salah et al. (2019)
Rice	Bags labeled with tags	No	Kumar and Iyengar (2017), Yakubu et al. (2022), Tao et al. (2021), Peng et al. (2022b) and Wang et al. (2022b)
Grain	Not identified	No	Thakur and Hurburgh (2009a), Gunasekera and Valenzuela (2020), Lucena et al. (2018) and Xu et al. (2022)
Cotton	An RFID or QR-code reader	No	Agrawal et al. (2021)
Coconut Oil	Integrate the identification of material flow with an code system	Yes	Gunawan et al. (2018)
Tobacco	GPS and RFID technologies	No	Chen et al. (2021)
Cocoa Bean	IoT and unique sensors	No	Perez et al. (2022)
Tea	RFID technologies	No	Liao and Xu (2019), Mao et al. (2021) and Paul et al. (2022)
Coffee	Packaged into bags with transaction code	No	Alamsyah et al. (2023), Dionysis et al. (2022) and Miatton and Amado (2020)
Olive Oil	IoT sensors dedicated to EVOO quality control	No	Arena et al.
Sugar	RFID and WSN	No	Ekawati et al. (2023)

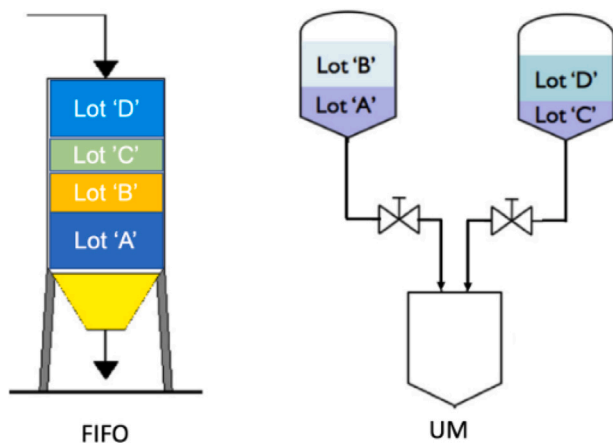


Fig. 4. The illustration of FIFO and UM agricultural product blending model adopted from Comba et al. (2013).

This can be done through the use of a identification management system (i.e., identification modeling) that records the movement of the product and provides information about its origin, composition, and handling, thereby improving the efficiency and accuracy of the traceability system.

In this sense, tracking bulk product on the supply chain, identification needs to be managed throughout the supply chain, from the field to the end consumer, to ensure the minimum lot id lost. Actually, even for non-bulk products, traceability can get disconnected due to identification inconsistency or loss during various stages of processing or transportation. A potential solution is the development of a robust digital identification model. If a discrepancy or inconsistency in identification arises, AI algorithms can scan the blockchain for the last known accurate data point, cross-referencing it with related batches to “rescue” or reconstruct the most likely traceability pathway.

5. Blockchain data traceability

A blockchain is a distributed, or decentralized, ledger—a digital system for recording transactions among multiple parties in a verifiable, tamperproof way. Applying blockchain for data traceability well satisfies the following traceability requirements: The flexibility of tracing back (supplier traceability) and tracking forward (client traceability); the immutability of data tracking across different stakeholders, because that the unique decentralized storage of blockchain technology does not rely on any organization or individual, and uses trusted technical means to publicly record all information on the “public ledger”; the data security of the traceability system, because that the data on the chain has a time stamp and cannot be tampered with is established; the degree of automation, since blockchain smart contract autonomously execute when predefined rules and conditions are met, which minimizes human intervention.

Permissioned blockchains, also known as consortium blockchains, are private networks that are restricted to a specific group of participants who are authorized to validate transactions and add blocks to the chain. Permissionless blockchains, also known as public blockchains, are open networks where anyone can participate and validate transactions. Examples of permissionless blockchains include Bitcoin and Ethereum. In a permissionless blockchain, anyone can join the network, validate transactions, and earn rewards for doing so. Thus, permissioned blockchains offer better security, privacy, and control through restricted access, streamlined governance, and scalability, making them more suitable for agriculture industries requiring a higher degree of trust and regulatory compliance.

5.1. Participants agreement

For agricultural product traceability, permissioned blockchains are preferred choice as compared to permissionless blockchains. This is due to the fact that permissioned blockchains offer greater control over participants and their activities, making them more suitable for enterprise-level applications such as supply chain management. To accomplish this, permissioned blockchains rely on participants agreement

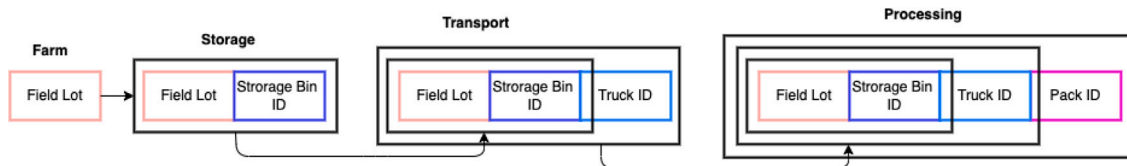


Fig. 5. An example of recording and linking up the identity across different stages of the supply chain.

to determine who can access the network, validate transactions, and update the ledger.

In a permissioned blockchain, the identity of each participant is known and authenticated. Participant agreements define the rules and protocols that govern how participants interact with the blockchain network. The terms of the agreement are directly written into lines of code in smart contracts, which are stored on the blockchain and automatically execute when certain conditions are met (Shrimali and Patel, 2022).

In general, a typical blockchain participants agreement covers: (1) Roles and Responsibilities: The agreement should define the roles and responsibilities of each participant in the supply chain. For example, it might specify that farmers are responsible for entering data on crop yields, while distributors are responsible for tracking shipping and handling information; (2) Data Sharing: The agreement should establish what and how data will be shared among participants, including the types of data that will be recorded on the blockchain and who will have access to it. This might include requirements for data privacy and security, as well as procedures for resolving disputes; (3) Verification and Validation: The agreement should specify how data will be verified and validated to ensure its accuracy and authenticity. This might include requirements for third-party audits, independent verification, or other forms of quality control; (4) Access and Permissions: The agreement should define who has access to the blockchain network and what permissions they have. This might include requirements for participant identity verification, as well as rules for revoking or updating permissions; (5) Smart Contracts: The agreement should outline the use of smart contracts, which are self-executing programs that automate transactions and processes on the blockchain. This might include specifications for the types of smart contracts that will be used, as well as requirements for their implementation and testing; (6) Governance: The agreement should establish rules for how the blockchain network will be governed and managed. This might include requirements for consensus mechanisms, decision-making procedures, and dispute resolution processes.

5.2. Real-time in-chain data

To create an effective traceability system for agricultural products, both the digital sensing and the blockchain platform need to handle the unique challenges that come with managing large quantities of goods that may be merged or split during the supply chain process. The integration of blockchain, IoT, and AI technologies can significantly improve product identification verification during the supply chain process (Deebak et al., 2022). Blockchain ensures secure and transparent data storage, while IoT devices capture and transmit real-time information. AI algorithms analyze and optimize the data to support decision-making, creating a seamless and reliable identification process throughout the entire supply chain. Fig. 6 illustrates the process of blockchain IoT real-time product identification verification, showcasing the synergy of these technologies and their critical roles in maintaining traceability for bulk agricultural products. Measure the time it takes for data from IoT sensors to be confirmed and recorded on the blockchain. Low latency and fast confirmation times are essential for real-time traceability. In practices, the key factors determines the latency of real-time in-chain data include:

- **Network Architecture:** The physical architecture of the blockchain network, including the number of nodes, their geographic distribution, and the quality of the network connections, directly affects latency. Networks with well-distributed nodes and low-latency connections can reduce propagation delays.
- **Block Time:** Block time is the time taken to create a new block on the blockchain. In networks with shorter block times, transactions are confirmed and added to the blockchain more frequently, resulting in lower latency.
- **Consensus Mechanism:** The consensus mechanism used by the blockchain impacts how quickly transactions are validated and added to the blockchain.

5.3. Unique for agricultural product traceability

It is common for blockchain technology to be applied to product traceability in various industries. The unique features of blockchain technology in agricultural products are influenced by the distinct characteristics of the agricultural supply chain and the specific regulations governing the industry. While the underlying technology and blockchain principles remain consistent across different applications, the behavior and implementation at the application level can vary significantly due to the nuances of each industry.

The nature of the agricultural supply chain, with its multiple stages (i.e., from farm to market/export) diverse stakeholders, and quality control requirements, results in specific use cases and data requirements that may differ from other industries. Compliance with agricultural regulations, such as food safety standards or organic certifications, also drives the need for tailored blockchain solutions. Please refer to the discovery on this in Section 7.

5.4. Accessing blockchain operation

The efficiency of blockchain operation is primarily impacted by the chain architectures. Multi-chain architectures allow parallel processing of transactions across multiple chains, which improves scalability and reduces latency. This structure enables different chains to handle specific tasks or types of transactions, leading to more efficient data management and increased throughput. For instance, a dedicated chain could manage supply logistics while another handles compliance and regulatory information, thereby optimizing resource allocation and improving overall system efficiency (Peng et al., 2022a). Heterogeneous chain architectures integrate various blockchain protocols within a single system. This setup allows different chains, each with its unique consensus mechanisms and security features, to coexist and communicate effectively. As such, heterogeneous chains can adapt to the specific needs of different stakeholders in the supply chain, offering tailored solutions that improve traceability and data integrity across diverse agricultural contexts. For example, in an agricultural context, a blockchain network might integrate a protocol optimized for rapid data logging from IoT devices monitoring soil conditions, alongside another protocol designed for secure, verified transactions between farmers and distributors. This combination enables tailored solutions that improve traceability and data integrity across diverse agricultural contexts (Kuzlu et al., 2019; Patelli and Mandrioli, 2020).

Technically, the performance metrics of blockchain operation can be summarized as the following key factors (Fan et al., 2020b; Patelli and Mandrioli, 2020):

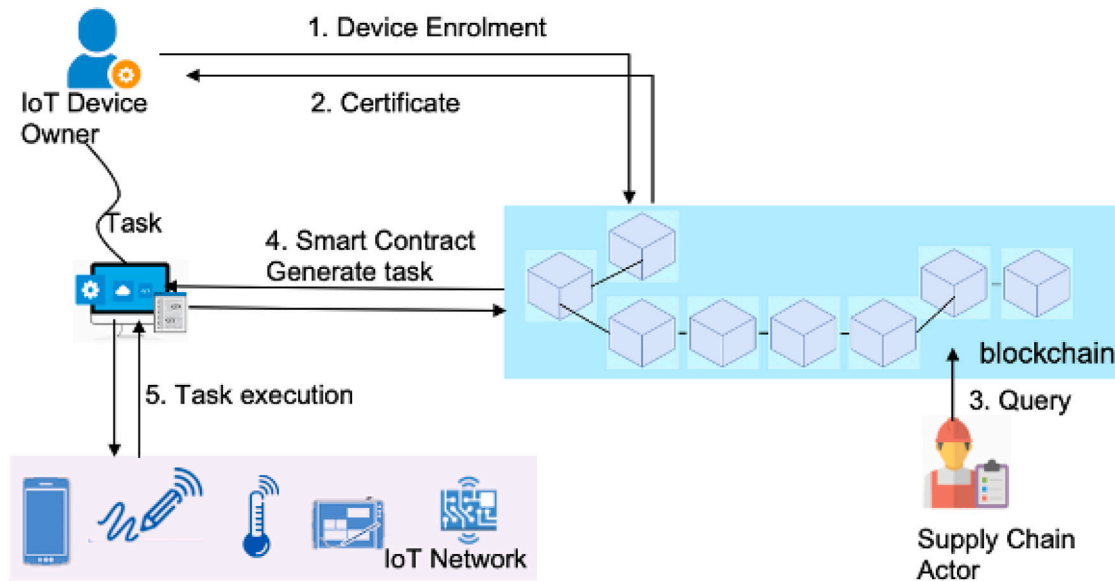


Fig. 6. The process of real-time product identification and in-chain data.

- **Data Integrity:** Assess the system's ability to ensure that the data remains unaltered and tamper-proof throughout its life cycle.
- **Data Security:** Evaluate the system's ability to protect data from unauthorized access or manipulation.
- **Data Privacy:** Determine the degree to which the system preserves data privacy for supply chain actors.
- **Transaction throughput:** The number of transactions that a blockchain network can process per second (Kuzlu et al., 2019).
- **Block size and interval:** The block size and interval determine the rate at which transactions are processed on the blockchain.
- **Latency:** This measures the time it takes for a transaction to be processed and validated on the blockchain (Kuzlu et al., 2019).
- **Scalability:** The ability of a blockchain network to handle increasing transaction volumes without impacting its performance.
- **Consensus Mechanism Efficiency:** Analyze the efficiency and security of the consensus mechanism used in blockchain (e.g., PoW, PoS, or Delegated PoS).
- **Security:** The security of a blockchain network is measured by the network's resistance to attacks, such as double-spending or 51% attacks.

To ensure the utmost efficiency and reliability in blockchain-based data traceability systems, these metrics must be carefully considered and regularly evaluated. As advancements in blockchain technology continue, it is essential for stakeholders to stay updated, ensuring that the chosen systems remain resilient, efficient, and attuned to the intricacies of data traceability needs. Regular reviews and adjustments, based on these key factors, will enable organizations to optimize blockchain data traceability, while safeguarding the integrity and security of the data.

Note that for traceability in bulk agricultural products, because the blending can happen at any stage of the supply chain, thus it is not possible to use/track one unique product identifiers, such as barcodes, RFID tags, or QR codes. All parties have to agree on a blending management model (to prevent inconsistent product identification) and quality estimation approach in case of blending.

Table 4 presents 16 blockchain traceability applications of 9 bulk agricultural products including rice, grain, tea, tobacco, cotton, coffee, sugar, palm oil, and soybean, where different blockchain technologies have been employed across these studies, with Hyperledger Fabric, Ethereum, and Parallel Blockchain being the most commonly used platforms. In our assessment of each application, we scrutinize their

performance and operational aspects through the lens of six critical criteria: Security and Trust (ST), Latency and Throughput (LT), the Degree of Data Integration (DI), Interoperability to ERP (IE), Permission Agreement (PA), and Compliance and Standardization (CS). The criteria of SP, LT, and DI primarily focus on the performance of the blockchain system itself, assessing its ability to maintain data security, handle transaction speed, and seamlessly integrate data, which will impact transaction throughput. In contrast, criteria IE, PA, and CS delve into the operational and deployment aspects of blockchain, which will be further explored in Sections 6 and 7.

As seen, each of these blockchain applications focuses on different aspects of the traceability system, reflecting the specific requirements and challenges of their respective agricultural supply chains. While the majority of the works prioritize aspects Security and Trust (ST) and Latency and Throughput (LT), it is worth highlighting that none of the examined applications encompass all six evaluation criteria. This observation underscores the inherent complexity of implementing blockchain traceability solutions in the agricultural sector.

It also highlights the need for further research and development to create comprehensive blockchain systems that address all aspects of traceability, ensuring not only data security and efficiency but also seamless integration, interoperability with existing systems, adherence to permissions, and compliance with industry standards.

6. Agricultural traceability system

An agricultural traceability system (TS) in place that allows grower to track the produce from the field to the consumer. The system can be queried to retrieve information related to incoming, internal and outgoing lots and to retrieve information that connects the individual incoming agricultural product lots to an outgoing shipment.

When deciding on TS adoption, it is crucial to discover the specialized field of the TS supplier which can provide insight knowledge and address the unique challenges and requirements of the business. In this context, the digital sensing capability of the TS supplier is essential because that different product involves different sets of sensor, and deployment of those sensors require specific domain knowledge. For bulk products, traceability disconnection may occur at any time during the supply chain process, which should be addressed either through identification modeling or by performing identity verification using Blockchain IoT and AI technologies. It is worth noting that blockchain AI holds significant potential for tracking the movement of agricultural

Table 4

Assessment of blockchain traceability for 9 bulk agricultural products across key criteria: Latency & Throughput (LT), Security & Trust (ST), Degree of Data Integration (DI), Interoperability to ERP (IE), Permission Agreement (PA), and Compliance & Standardization (CS).

Work	Blockchain	Product	LT	ST	DI	IE	PA	CS	Focus of this work
Yakubu et al. (2022)	Ethereum	Rice	✓	✓	×	×	×	✓	Sequential diagram of supply chain process
Tao et al. (2021)	Hyperledger Fabric	Rice	×	✓	×	×	×	✓	Traceability system architecture and sequence diagram
Peng et al. (2022b)	Parallel blockchain	Rice	✓	✓	×	×	×	×	Cross-chain model of rice supply chain supervision
Wang et al. (2022b)	Hyperledger fabric	Rice	✓	✓	×	×	●	×	Supply chain information supervision model
Kumar and Iyengar (2017)	Simulation	Rice	×	×	×	×	×	×	Blockchain framework for supply chain management
Gunasekera and Valenzuela (2020)	Simulation	Grain	×	✓	×	×	×	×	Productivity gain due to blockchain adoption
Lucena et al. (2018)	Hyperledger fabric	Grain	×	✓	×	×	✓	✓	Implementation of Blockchain for grain quality assurance tracking
Xu et al. (2022)	Hyperledger fabric	Grain	✓	✓	✓	×	×	×	Reliable traceability prototype system for wheat quality and safety
Liao and Xu (2019)	Hyperledger fabric	Tea	✓	×	×	×	×	×	Blockchain system for tea quality traceability
Mao et al. (2021)	Ethereum	Tea	✓	×	×	×	×	×	Tea traceability consensus mechanism
Chen et al. (2021)	Hyperledger fabric	Tobacco	×	✓	×	×	×	✓	Blockchain-based traceable and verifiable logistics system
Agrawal et al. (2021)	Simulation	Cotton	✓	×	×	×	×	✓	Blockchain framework for cotton supply chain traceability
Miatton and Amado (2020)	Hyperledger fabric	Coffee	×	×	×	×	×	✓	Blockchain based global coffee value chain
Ekawati et al. (2023)	Ethereum	Sugar	✓	×	✓	×	×	✓	Blockchain based white crystal sugar agro-industrial supply chain system
Hirbli (2018)	Not identified	Palm oil	✓	×	✓	✓	✓	×	Blockchain based palm oil supply chain traceability
Salah et al. (2019)	Ethereum	Soybean	×	✓	✓	×	×	✓	Data storage, smart contract and sequential diagram of supply chain process

(✓): Supported; (×): Not Supported; and (●): Partially Supported.

Table 5

The list of commercial traceability systems on the market.

Name	Agricultural products	Country	Technology	URL
Provenance	Food, fashion, and beauty	UK	Blockchain	provenance.io
Ambrosus	Food & pharmaceutical	Swiss	Ethereum+IoT, realtime tracking	ambrosus.io
Ripe.IO	Food	US	Blockchain+IoT	ripe.io
IBM Food Trust	Food	US	IBM Hyperledger	hyperledger.org
TE-FOOD	Food	Vietnam	TE-FOOD Blockchain+IoT+AI	te-food.com
VeChain	Fashion, luxury goods, and agriculture	China	VeChainThor+IoT realtime tracking	vechain.org
Aglive	Livestock and meat products	Australia	Blockchain+IoT+AI realtime tracking	aglive.com
TraceHarvest	agricultural Product	US	Proprietary platform	blockapps.net
GrainDiscovery	Grain	Canada	Blockchain and satellite imagery	graindiscovery.com
Origintrail	Food and beverages	Slovenia	Blockchain data sharing	origintrail.io
TUV SUD	Goods and services	Germany	Blockchain+IoT+AI	tuvsud.com

products throughout the supply chain ([Qian et al., 2022b, 2020](#)). This technology not only can aid in product identification upon traceability disconnection, but also responds to supply chain issues in real-time, such as delays or disruptions. [Table 5](#) summarizes the list of blockchain based agricultural TS available in the market.

In the evaluation of a blockchain-based agricultural product traceability system (TS), the assessment primarily hinges on the following factors:

- **Security:** Even with blockchain's decentralized framework and cryptographic protections that deter tampering and fraud, it remains crucial to rigorously evaluate its security mechanisms to ensure data integrity and resistance to potential breaches, because if a security system is not deployed, operated, and managed properly, it can expose vulnerabilities, rendering the entire traceability chain ineffective and susceptible to malicious activities.
- **Accuracy:** The immutability of blockchain ensures the accuracy of data. Once data is recorded on the blockchain, it cannot be changed, which ensures the reliability of the information. However, inaccuracy might come from digital sensing during data

capture, thus its vital to ensure the TS is equipped with a data validation model for consistent product identification

- **Speed and Ease of Use:** How quickly and easily customers can access traceability information also plays a crucial role in satisfaction. The user interface of the application used to access this information should be intuitive and user-friendly.
- **Customer Support:** Good customer service is important for addressing any issues or questions that customers may have, and it can significantly impact overall satisfaction.
- **Interoperability to ERP:** The integration capability with existing ERP systems is a critical factor, especially for ensuring a smooth transition and alignment with existing enterprise workflows. We discuss the next its significance for industry adoption.

6.1. Interoperability to ERP

ERP (Enterprise Resource Planning) software ([Banerjee, 2018](#)) has been widely used by companies in the Agriculture supply chain to

Table 6

The list of Australian Regulations/Guidelines for Grain Supply Chain Traceability.

Regulation/Guideline	Publisher	Focus
Management of Grain Within the Australian Grain Supply Chain, Australian Grain Industry – Code of Practice	Grain Trade Australia (GTA)	Production, Storage, Transport and Export
Grain Industry Transport Code of Practice	Grain Trade Australia (GTA)	Transport
Farm Biosecurity Manual for the Grains Industry	Grain Producers Australia (GPA)	Production
Agricultural Export Price and Volume Indicators	Australian Department of Agriculture and Water Resources (DAWR)	Export
Measuring Chain Volumes for Exports of Goods & Services	Australian Bureau of Statistics (ABS)	Export
Australia's Grain Supply Chains	Grains Research and Development Corporation (GRDC)	Production to market
Understanding Australian Wheat Quality	Grains Research and Development Corporation (GRDC)	Production and processing
National Food Traceability Implementation Framework	GS1 Australia	Production to market
IEEE Standard for Framework of Blockchain-based Internet of Things (IoT) Data Management	IEEE	Blockchain Data Management

manage various business processes, including procurement, inventory management, production planning, and distribution. For example, an ERP for farming can help farmers in tracking which crop is grown in a particular field. In practice, these systems generate a lot of data that is critical to digital traceability of the supply chain. The interoperability to ERP defines the system's capability to seamlessly exchange and utilize data with existing ERP systems. This determines how easily the blockchain-based TS can interact with other supply chain systems and processes, as well as other blockchain networks.

By interacting with existing ERP software, companies can ensure that the ERP system (internal traceability) is seamlessly integrated with the data generated by external traceability which are mainly stored on the cloud. This will enhance agricultural product traceability, meanwhile ensuring compliance with regulatory standards. As such, integrating blockchain-based TS with existing ERP software can also help companies to improve supply chain transparency, reduce the risk of fraud, and enhance customer trust. From an industry adoption perspective, interoperability to ERP facilitates the transition from traditional ERP-centric systems to next-generation blockchain-based digital traceability. This seamless integration ensures that businesses can evolve without disrupting existing workflows and data structures.

7. Compliance and standardization

7.1. Compliance

Agricultural traceability compliance pertains to the adherence of businesses within the agricultural sector to established regulations and guidelines that ensure the transparency and accountability of their supply chains. Such compliance standards are integral to maintaining food safety, consumer trust, and facilitating international trade (Corallo et al., 2020). These regulatory frameworks, often country-specific, provide a structured approach to tracking and verifying the journey of agricultural products from their origin to the end consumer.

Adhering to these regulations and guidelines is crucial for businesses in the agricultural sector to comply with traceability requirements (Thakur and Hurburgh, 2009b). The standardization provided by these regulations and guidelines offers a clear and widely-accepted framework that increases the likelihood of meeting regulatory standards. By following these standards, businesses can enhance transparency, accountability, and ultimately maintain consumer confidence in the supply chain. The agricultural industries in different countries have specific regulations and guidelines in place to ensure traceability within their supply chains. For example, the European Union's Digital Product Passport is a standard for product information and traceability

across member states, which ensures alignment with EU legislation and facilitates compliance with its traceability requirements. As a case in point, Australia, has strict regulations and guidelines for grain supply chain traceability. These regulations are listed in Table 6, which encompasses the entire grain supply chain, from production to export or the market.

7.2. Standardization

Standardization of traceability stands as a pivotal future direction, given its potential to bridge disparate traceability systems, IoT devices (digital sensing), and blockchain networks. This interoperability is vital for creating a unified and efficient traceability network, allowing data to be easily shared and understood globally by all supply chain actors.

Standardization on digital sensing for agricultural product traceability involves establishing a set of unified guidelines for data collection, transmission, and processing across various IoT devices and sensors. Table 7 lists the key standardization items for digital sensing in agricultural product traceability. These guidelines should regulate the behavior of digital sensing throughout the entire production process, ensuring consistent and accurate data collection for large quantities of goods. Furthermore, standardization should also set rules for handling cases of errors or inconsistent product identification in the case of bulk agricultural products, providing clear procedures for identifying and rectifying such issues. This is particularly important when dealing with the unique challenges of managing large quantities of goods that may be merged or split during the supply chain process.

Standardization on blockchain data traceability can be achieved by establishing a common set of protocols, data formats, and methodologies to ensure seamless interaction and data sharing between different blockchain networks and traceability systems (Wang et al., 2022a). This involves defining (1) standardized data for product quality control and assessment. For example, for grain quality (grade) authentication on blockchain, some common data that are required include: grain variety and type, location and date of production, and agricultural practices which include information on the specific agricultural practices used to grow the grain, such as the type of seeds used, the type of soil, the use of pesticides or fertilizers, and the record of grain transport, storage and processing; (2) standardized product characteristic measurements. For example, in measuring grain quality in Australia, we measure protein level, starch content, test weight, moisture content, contaminant, falling number, and kernel color; and (3) Access standards for stakeholders joining the blockchain-based traceability platform. Table 8 outlines key standardization elements in permission agreements for blockchain-based agricultural traceability.

Table 7
Key standardization items for digital sensing in agricultural product traceability.

Item	Description
Sensors used	Specifying the types and models of sensors approved for use, ensuring quality and consistency in data capture.
Sensor calibration	Establishing standard procedures for calibrating sensors to ensure accurate data capture across devices and deployments.
Data verification	Outlining methods and techniques to verify the authenticity and accuracy of the data captured.
Data recording	Setting protocols for how data is to be recorded, whether it is locally on the device, sent to a cloud, or stored in a database. Ensuring data integrity and timestamping.
Sensor placement and use	Providing guidelines on where and how sensors should be placed on agricultural products to achieve the most accurate readings.
Data transmission	Standardizing the methods, protocols, and frequencies for transmitting data from sensors to storage or analysis platforms.
TRU	Defining the smallest unit within a supply chain that is tracked or traced.

Table 8
Key standardization items in permission agreements for blockchain-based agricultural traceability.

Section	Description	Key elements
Access control	Define who can access the blockchain and at what level.	User roles and permissions Verification processes Access levels
Data management	Outlines protocols for data handling on the blockchain.	Data entry, storage, and retrieval Data standards and integrity Privacy measures
Quality assurance and compliance	Ensures grain quality meets established standards.	Grain classification and standards Inspection and certification processes Compliance with industry regulations
Security	Details security measures to protect the blockchain.	Encryption standards Security audits Incident response plans
Dispute resolution	Defines how disputes are resolved within the network.	Dispute resolution processes Arbitration mechanisms
Legal and regulatory compliance	Ensures compliance with laws and regulations.	Adherence to legal frameworks Regular compliance reviews

8. Industry adoption

8.1. Pathway

The adoption of agriculture traceability systems is a key driver of supply chain digital transformation. Fig. 7 presents a pathway to agricultural product traceability implementation with key control points to ensure successful traceability deployment and operation.

In planning a specific agricultural product traceability, it is essential to ensure full coverage of the entire supply chain from farm to market. The objective of implementing traceability may vary depending on the intended use case, ranging from product quality assurance (Hassoun et al., 2023a) to fully automated trade capabilities. Note that these varying objectives entail different levels of investment and associated risks. In practice, it is advisable to initially focus on establishing traceability for quality assurance purposes and subsequently progress towards integrating automated trade functionalities. This approach allows for a more measured and scalable implementation of the traceability system.

In designing a traceability system, inconsistent identification is the primary cause of traceability disconnection. Therefore, IoT sensing capabilities, particularly those tailored to specific products, serve as a crucial mitigation strategy. Implementing effective identification management to address traceability disconnections ensures a more reliable and seamless traceability system. The options for data traceability platforms encompass centralized ERP systems, permissioned blockchains, and permissionless blockchains. Centralized ERP systems may not provide the electronic trust required for seamless collaboration and data sharing among the various parties involved in the supply chain. It is recommended to begin with permissioned blockchains, as

they offer a more controlled environment, and later consider scaling to permissionless blockchains to target future public data traceability requirements.

It is worth noting that traceability protocols play a critical role in determining the compatibility of traceability system implementation. These protocols encompass compliance with national and international standards, as well as the standardization of traceability practices, enabling the merging of different traceability blockchains. While traceability compliance has been largely addressed (Thakur and Hurburgh, 2009b; Corallo et al., 2020), traceability standardization is still a work in progress and merits special consideration, particularly for traceability systems associated with exports.

8.2. Strategies to accelerating industry adoption

The adoption of agricultural product traceability presents significant challenges, primarily due to the inherent complexity of the agricultural supply chain and the difficulty in coordinating all parties involved. However, it has been demonstrated that large buyers in the market can effectively drive the implementation of traceability systems. For instance, Walmart successfully championed the adoption of IBM's Food Trust. BYD, one of the world's largest electric car manufacturers, collaborated with VeChain and DNV GL (a global quality assurance and risk management company) developed a carbon credit solution on the VeChainThor blockchain (BYD, 2018). The success can be attributed to the fact that large buyers have the resources and influence to set traceability and data sharing standards and to invest in and implement blockchain-based systems. These buyers can require suppliers, including SMEs, to join the platform and share data about their products while also helping them cover the costs of adopting such systems. This helps

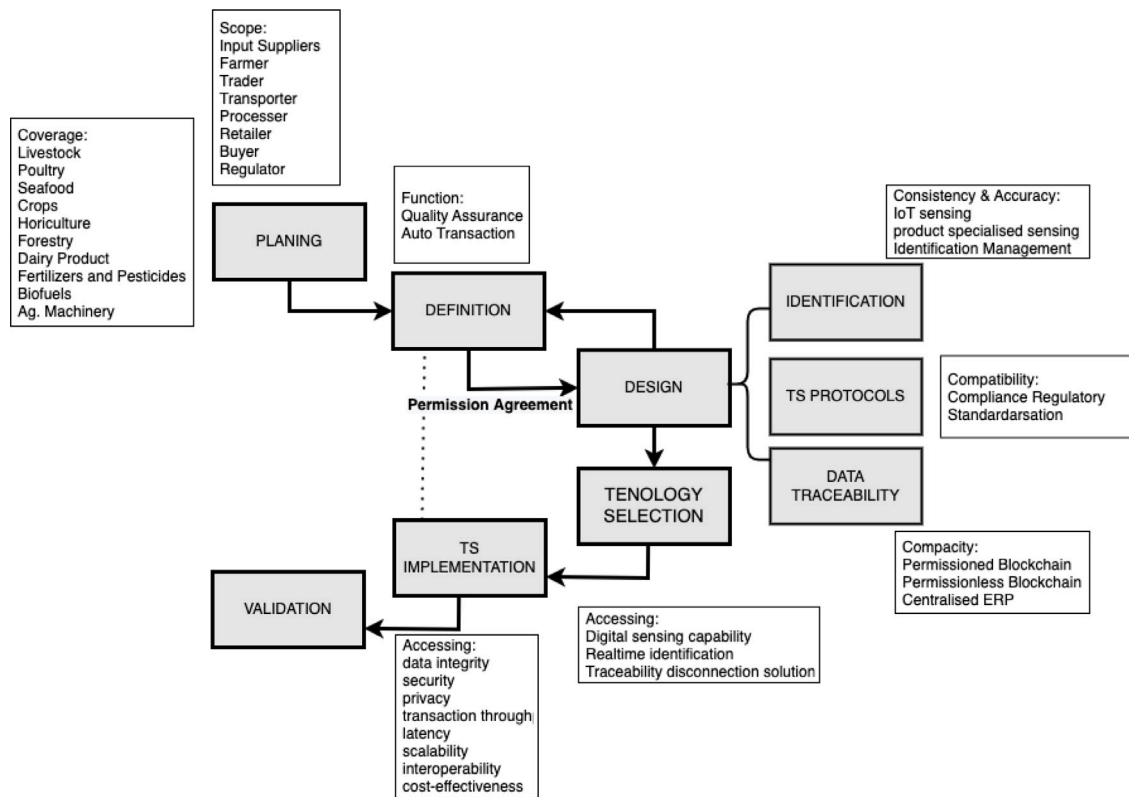


Fig. 7. A pathway to adopting agricultural product traceability with key performance assessment points listed at each step.

overcome the cost constrain which facilitates SMEs in the adoption of blockchain-based product traceability solutions.

Alternatively, government involvement is crucial for the successful adoption of blockchain applications in the agricultural sector. They have the authority to set standards, promote transparency, and ensure food safety and product authenticity. By facilitating collaborations between tech innovators, agricultural stakeholders, and research entities, governments can also drive technological advancements and ensure its practical implementation. For instance, in 2018, in alliance with Alibaba, the government rolled out a pilot initiative aimed at tracing and verifying the origins of imported goods through blockchain, reinforcing the integrity and safety of agricultural goods entering the Chinese market. A year later, Walmart China and VeChain teamed up to create a traceability platform grounded in the VeChainThor Blockchain.

More specifically, the government can expedite standardization of the blockchain-based traceability system. This involves implementing regulatory frameworks that standardize data input formats and interoperability guidelines, ensuring that different traceability systems can interact seamlessly and communicate with one another, which in turn fosters widespread compatibility and adoption. They can also invest blockchain traceability initiatives, and incentivize stakeholders along the same supply chain to form the alliance of blockchain traceability. For example, they could do this by implementing financial or tax policies that reward the integration of blockchain for quality traceability and automated trading. These incentives could take the form of tax breaks, grants, or subsidies for those who choose to employ these systems. By taking these strategies, the government can facilitate the wide-scale adoption of blockchain-based traceability. Note that such government-led action can potentially be even more influential than the efforts made by major buyers in the industry.

From the view point of business owner, to accelerate the adoption of agricultural product traceability, especially for SMEs they should collaborate with industry partners and engage with other stakeholders in the supply chain to establish a shared vision for implementing

traceability systems and the benefits it can bring to all parties. In particular, partnering with influential market players or participating in government-driven blockchain traceability programs can help leverage their resources, authority, and expertise in the field, ensuring a more effective and widespread adoption of traceability solutions across the supply chain.

Nevertheless, given the high costs and complexity associated with implementing BAT systems, small-scale farmers and those in regions with limited infrastructure might benefit from alternative techniques. ACID-compliant databases can serve as a reliable option, ensuring data integrity and security through centralized systems that are easier and cheaper to implement than blockchain. Time series databases offer another effective solution, particularly for tracking and monitoring agricultural activities over time due to their efficiency in handling time-stamped data and real-time analytics capabilities. Additionally, cloud-based traceability systems provide scalability and flexibility at reduced upfront costs, making them accessible from various devices and locations. RFID and barcode systems remain proven, low-cost technologies that simplify the tracking of physical goods without the complexity of blockchain. These alternatives can provide robust traceability solutions, enhancing data management and traceability while being more economically viable for small-scale farmers.

9. Future directions

In the efforts to developing full-capacity digital agricultural traceability, the future works are concluded as follows.

9.1. Decipher the agricultural plant lifecycle

As most existing agricultural traceability focus on supply chain traceability (external) and/or internal agricultural production traceability. As discussed in Section 2.3, they typically follow the sequence of operations such as seeding, watering, fertilizing, and other farm

management activities. While these aspects are crucial for maintaining product quality and safety, they often overlook the detailed traceability of the agricultural plant lifecycle itself. This lifecycle traceability includes critical stages such as germination, growth, flowering, fruiting, and harvesting, which are essential for understanding the complete history and quality of the agricultural products. Therefore, it is necessary to decipher the agricultural plant lifecycle for digital traceability, enabling a more comprehensive and accurate traceability system. This approach not only enhances transparency and accountability but also ensures that every phase of the plant's development is documented and traceable, thus improving overall supply chain integrity and consumer trust.

9.2. Digitize agricultural traceability compliance

The agricultural traceability compliance in Section 7.1 provides a clear and structured framework that is easier for businesses to understand and implement. This helps ensure that key regulatory requirements are met across major checkpoints, thereby maintaining a baseline of transparency and accountability. The review focus on the broader stages of the supply chain, not yet delving into specific activities and checking points. This is insufficient for achieving full capacity digital traceability, especially within the context of BAT. To truly leverage the potential of BAT, compliance must be digitized down to the granular level of checking points. This detailed digitization allows for the creation of smart contracts that can automatically enforce compliance by executing predefined rules and procedures. Moreover, This precision will also open the door for integrating AI technologies, providing real-time monitoring, analysis, and decision-making support. By addressing these gaps, agricultural businesses can achieve a more transparent and accountable supply chain, thereby enhancing consumer trust and meeting stringent regulatory requirements more effectively.

9.3. AI for blockchain-based agricultural traceability

As per AI has demonstrated super capability on computer vision, big data analytics, optimization and language understanding, integrating AI to BAT we recommend the following four aspects of traceability developments:

- AI for agricultural product quality check through image recognition: Inspects crops and products for defects, damage, or signs of disease using AI-powered image recognition technology.
- AI for automated compliance checks: Utilizes AI to understand and process text and procedural data, verifying adherence to regulatory standards at each stage of the supply chain.
- Supply chain optimization via big data analytics: Employs AI to analyze large datasets, forecast demand, detect anomalies, and recommend corrective actions to minimize transit times and maintain product freshness.
- Quality prediction via AI quality decay modeling: Uses AI to develop quality decay models that predict product quality based on factors such as environmental conditions and handling practices, enabling proactive quality management.

10. Conclusion

This paper delves into the realm of blockchain-based digital traceability for agricultural products, providing an extensive traceability evaluation review that revolves around two vital dimensions: digital sensing and blockchain data traceability, complemented by traceability protocols. In the exploration of digital sensing, the study brings attention to the intriguing challenge of inconsistent product identification, with a particular focus on its implications in the case of bulk agricultural products.

When examining blockchain data traceability, we emphasize the paramount importance of system compatibility and the need for standardization to ensure seamless integration. Through this meticulous review, We discuss and the challenges entailed in implementing blockchain-based digital traceability, also present an innovative and comprehensive technique assessment framework. By establishing a clear pathway for the adoption of traceability techniques, this research paves the way to traceability 4.0 b (Hassoun et al., 2023b), a future of heightened transparency, efficiency, and trust in the agricultural industry.

In order to practically adopt and implement blockchain-based agricultural traceability throughout the industry, we have the following recommendations:

- Implement a staged process, prioritizing quality assurance first before enabling automated trading.
- Address inconsistent product identification which often causes traceability disconnection not only by adopting advanced sensor technology, but also via traceability modeling and operation management.
- lack of interoperability is the biggest barrier of technique adoption. influential market players, such as big buyers, retailers, or government agencies, should act as coordinators to bring all stakeholders together and establish industry-wide initiatives.
- Move forward the standardization of digital traceability implementation in Agriculture industry. This effort will facilitate interoperability and enable efficient traceability implementation.

Looking ahead, we anticipate several noteworthy trends of agricultural traceability. Firstly, owing to advancement of IoT and digital sensing, we are witnessing a significant transition from sporadic product identification to the spatio-temporal whole-life-cycle product identification. Secondly, there will be a shift towards transitioning from private permissioned to public permissionless blockchain traceability, opening the door for “traceability as a service”. Lastly for traceability compliance and standardization, the trajectory is moving from localized, country-specific to a more cohesive global, fostering consistency and interoperability across borders. These emerging trends collectively signify the widespread adoption of blockchain-based agricultural product traceability, which will revolutionize the tracking, monitoring, and global trade of agricultural products.

CRedit authorship contribution statement

Shaoning Pang: Writing – original draft, Project administration, Investigation. **Shyh Wei Teng:** Writing – review & editing, Project administration, Funding acquisition. **Manzur Murshed:** Writing – review & editing. **Cuong Van Bui:** Writing – review & editing, Investigation. **Priyabrata Karmakar:** Writing – review & editing. **Yanyu Li:** Validation. **Hao Lin:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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